Python Programming Foundations

Week 5 - Files, Errors, & Data Persistence

Day 3: "Comparing Data Representations & Application Structure"

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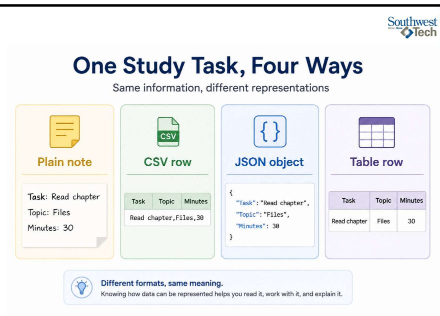
Same Information, Different Forms

- CSV shape helps with rows and columns.
- JSON shape helps with labels and nested information.
- Different shapes make different tasks easier.

2



Same Information, Different Forms

3

Review: Structure Data Has Shape

- CSV shape helps with rows and columns.
- JSON shape helps with labels and nested information.
- Different shapes make different tasks easier.

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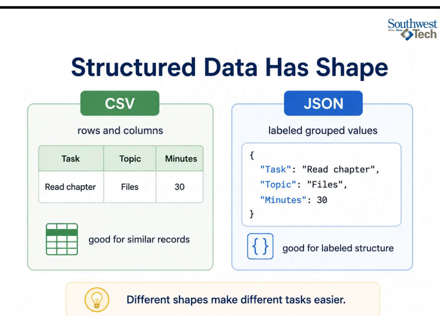

Shape of Data (Structured)

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What Counts as Success Today

- recognize at least two representations
- compare what each makes easier
- compare what each makes harder
- connect the representation to a larger application need
- avoid treating advanced structures as automatically better

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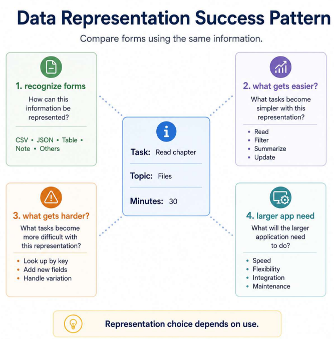

Today's Success

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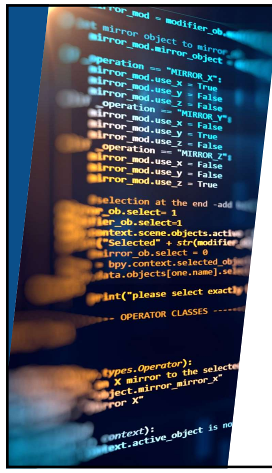

What We Will Use Today

- plain text
- CSV
- JSON
- list/dictionary structures
- table-like or model-like previews

We will ask what each form makes easier or harder.

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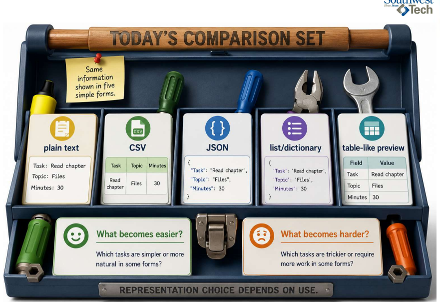

Toolbox

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What We Will "Skip" For Now

- Building a database
- Writing SQL
- Implementing an ORM
- Designing a production data model

Today is recognition and comparison.

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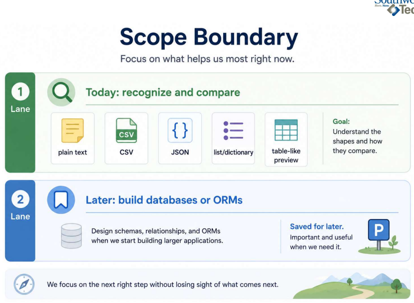

Parked For Later

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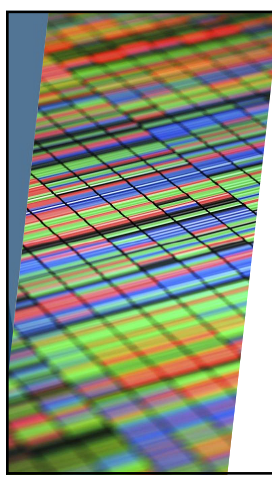

Data Forms

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Plain Text Is Human Friendly

- Plain text is easy for humans to read.
- It becomes harder when the program needs to reliably select fields, filter records, or compare values.

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Plain Text

Easy for humans

Task: Read chapter
Topic: Files

vs.

Harder for programs to select fields

Read chapter
Files

Plain text is useful, but structure is limited.

“Human” Friendly: Plain Text

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CSV Is Table Friendly

CSV works well when data fits rows and columns.

It is useful for lists of similar records, but it can get awkward when data needs deeper nested structure.

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CSV

Comma-Separated Values

Task	Topic	Minutes
Read chapter	Files	30
Review notes	Data shapes	20
Practice examples	Algorithms	45

Good for similar records
Works well when each row has the same fields.

Awkward for deeply nested data
Hard to represent complex or hierarchical relationships.

“Table” Friendly: CSV

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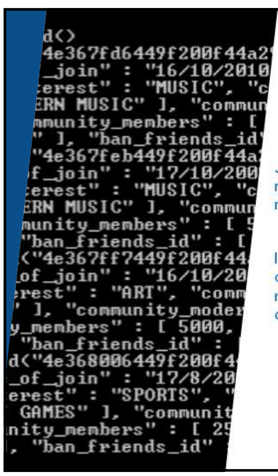

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JSON Is Structure Friendly

JSON works well when data needs labels, groups, or nested pieces.

It can be very useful, but it can also become hard to read if the nesting gets too deep.

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JSON

JavaScript Object Notation

```
{
  "task": "Read chapter",
  "topic": "Files",
  "minutes": 30
  "status": "planned",
  "details": { ... }
}
```

```
details
{
  "book": "Intro to Algorithms",
  "chapter": 2
  "type": "reading"
}
```

Good for labeled groups
Names make the data easy to understand and select.

Can get hard to read if nesting grows
More levels add detail but can reduce clarity.

“Structure” Friendly: JSON

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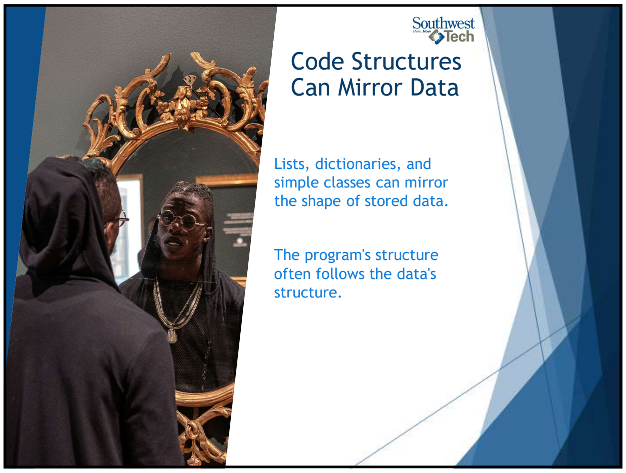

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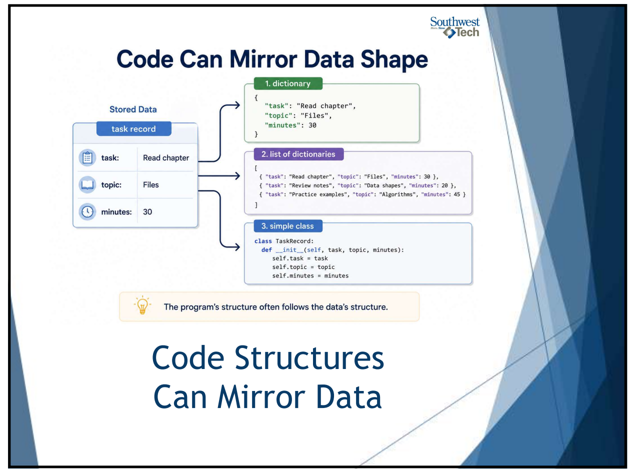

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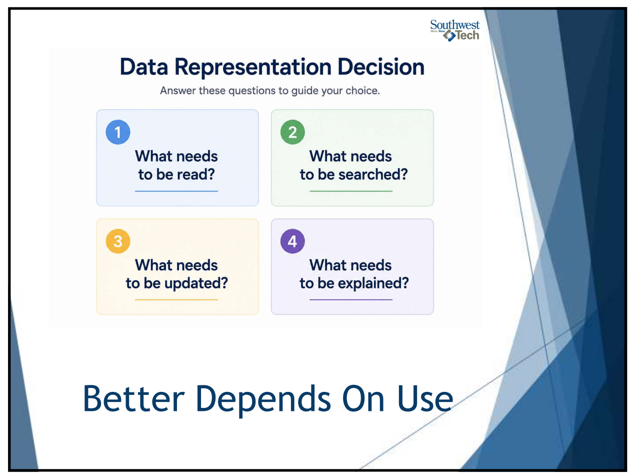

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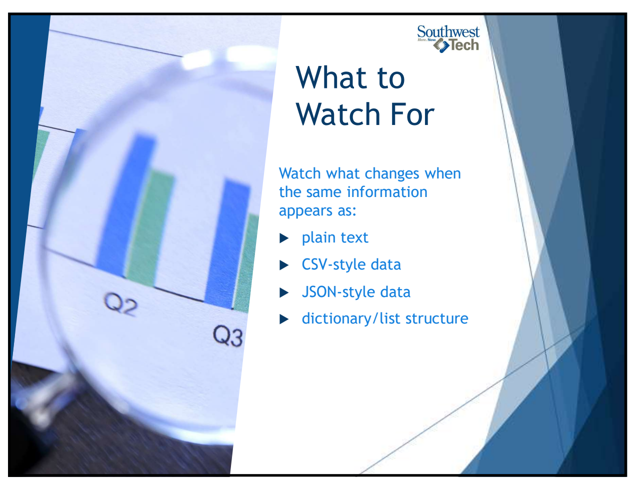

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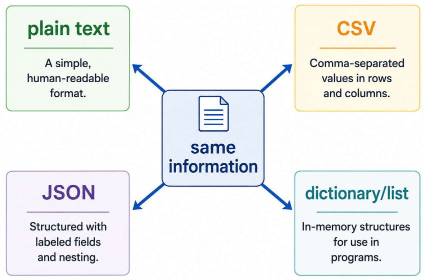

Demo: Same Data, Different Representations

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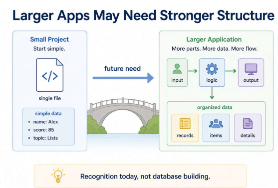

Preview: Larger Apps

As programs grow, data often needs more structure.

That does not mean we build a database today. It means we can recognize why larger applications need organized data.

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Common Failure: Bigger Is Not Automatically Better

A database is not automatically better than a file.

A class is not automatically better than a dictionary.

The representation should fit the problem.

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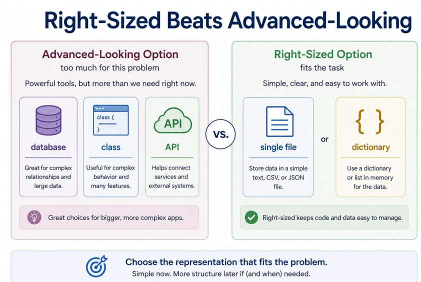

Common Failure: Bigger Is Not Automatically Better

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Assignment #10

For Assignment 10:

- Compare at least two representations of the same information.

Explain:

- What each form makes easier,
- What it makes harder,
- Where a larger application might need more structure.

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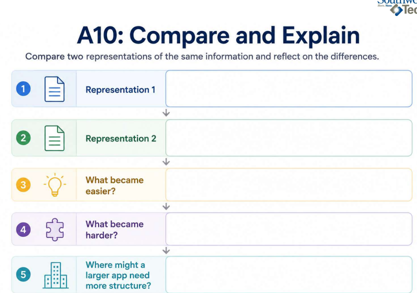

Assignment 10

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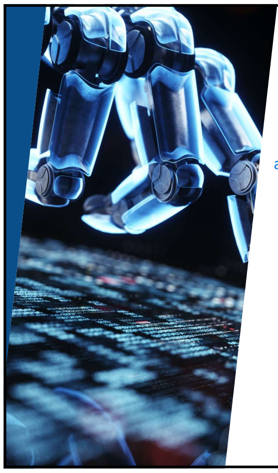

AI Use In Lab On

a

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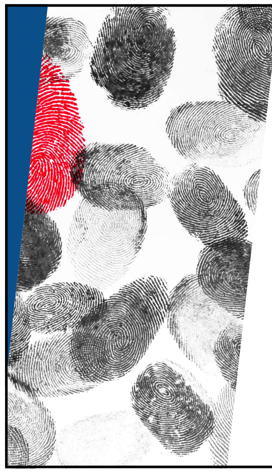

Evidence For A10

Your response should include:

- at least two representations
- clear easier/harder comparison
- beginner-level vocabulary
- connection to a larger application need
- AI-use note if AI helped explain a term

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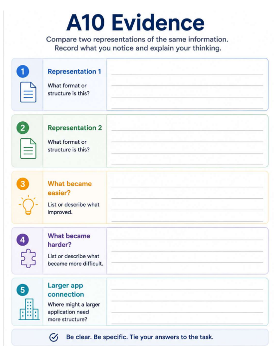

A10 Evidence

Compare two representations of the same information.
Record what you notice and explain your thinking.

1 Representation 1
What format or structure is this?

2 Representation 2
What format or structure is this?

3 What became easier?
List or describe what improved.

4 What became harder?
List or describe what became more difficult.

5 Larger app connection
Where might a larger application need more structure?

Be clear. Be specific. Tie your answers to the task.

Evidence

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Closing Success Check

By the end of today, you should be able to say:

"This representation is useful for this purpose, but another representation may be better for a different purpose."

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Thank you!

Have Fun!

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